

Quiz 1 – Instructor Key & Grading Rubric

Student Name: _____

Student ID: _____

Problem Setup

Define events explicitly (1 pt for labeling all correctly):

- L_1 : “Product is from Line 1”
- L_2 : “Product is from Line 2”
- L_3 : “Product is from Line 3”
- D : “Product is defective”

Given Data:

Line	Fraction of total output	Defect probability
L1	0.40	0.03
L2	0.35	0.05
L3	0.25	0.08

Question 1 (4 pts) – $P(D)$

Solution:

Use **law of total probability**:

$$P(D) = P(D | L_1)P(L_1) + P(D | L_2)P(L_2) + P(D | L_3)P(L_3)$$

Substitute numbers:

$$P(D) = (0.03)(0.40) + (0.05)(0.35) + (0.08)(0.25) = 0.012 + 0.0175 + 0.02 = 0.0495$$

Answer: $P(D) = 0.0495$

Points Allocation:

- 2 pt: Correct formula
 - 2 pt: Correct substitution
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Question 2 (4 pts) – $P(L_3 | D)$

Solution:

Use **Bayes' theorem**:

$$P(L_3 | D) = \frac{P(D|L_3)P(L_3)}{P(D)} = \frac{0.08 \cdot 0.25}{0.0495} \approx 0.404$$

Answer: $P(L_3 | D) \approx 0.404$

Points Allocation:

- 2 pt: Correct identification of Bayes' theorem
 - 2 pt: Correct substitution of numbers
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Question 3 (4 pts) – Defective items reaching customer

Solution:

- Only **undetected defective items** reach the customer.
- Detection rates: $L_3 = 0.75$, $L_1/L_2 = 0$.
- Probability defective reaches customer:
$$P(D_{\text{reach}}) = P(D | L_1)P(L_1) + P(D | L_2)P(L_2) + P(D | L_3)P(L_3)(1 - 0.75)$$
$$P(D_{\text{reach}}) = 0.03(0.40) + 0.05(0.35) + 0.08(0.25)(0.25) = 0.012 + 0.0175 + 0.005 = 0.0345$$
- Probability it came from L_3 given it reached customer:
$$P(L_3 | D_{\text{reach}}) = \frac{0.08 \cdot 0.25 \cdot 0.25}{0.0345} = \frac{0.005}{0.0345} \approx 0.145$$

Answer: $P(L_3 | D_{\text{reach}}) \approx 0.145$

Points Allocation:

- 1 pt: Correct adjustment for inspection system
 - 2 pt: Correct numerator/denominator setup (Bayes)
 - 1 pt: Correct substitution
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Question 4 (4 pts) – Mutually exclusive / independent

Solution:

- L_3 and D are **not mutually exclusive**, because some products from L_3 **can be defective**
 $(P(L_3 \cap D) = 0.08 \cdot 0.25 = 0.02)$
- Test independence:
 $P(L_3)P(D) = 0.25 \cdot 0.0495 \neq P(L_3 \cap D)$ — Not independent

Points Allocation:

- 1 pt: Correct answer for mutual exclusivity
 - 1 pt: Correct answer for independence
 - 1 pt: Supporting calculations/formulas
 - 1 pt: Clear explanation
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Question 5 (4 pts) – Same defect rate

Solution:

- If **all lines had same defect rate**, then
 $P(D | L_1) = P(D | L_2) = P(D | L_3) = k$
- Then $P(L_3 \cap D) = P(L_3)k = P(L_3)P(D)$
- **Independence holds**, because probability of defect no longer depends on line.

Answer: Yes, independent.

Points Allocation:

- 1 pt: Correct reasoning about equal defect rates
 - 1 pt: Correct use of independence formula
 - 1 pt: Clear explanation
 - 1 pt: Correct final conclusion
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